

Jackson Dowie

Nuclear Physicist

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🌐 [JacksonDowie.github.io](https://github.com/JacksonDowie)

🔗 [JacksonDowie](#)

Objective

Passionate experimental nuclear physicist building a career in nuclear physics, aiming to advance the field through comprehensive, end-to-end research. Committed to driving forward the development of nuclear technology and applications, and fostering innovation in experimental approaches and methodologies.

Employment

2025 – Jan **AMS Specialist Programme Support Officer**, *Australian National University*, Canberra

Accelerator mass spectrometry of Cl-36, development of gas ionization detectors, analogue and digital data acquisition systems, along with analytical chemistry and sample preparation, external user support, and business development and outreach.

2023 – 2024 Mar Feb **Postdoctoral Research Scholar**, *University of Kentucky*, Lexington

γ -ray and fast-neutron spectroscopy using HPGe and liquid scintillator detectors. Ran the particle accelerator and laboratory, built and maintained the electronics system, drafted publications and presented work at conferences and laboratories.

2022 – 2023 Aug Feb **ANU Research Officer**, *Australian National University*, Canberra

Physicist at the Department of Nuclear Physics at the ANU. Wrote documentation and manuals for the new space radiation testing beamline.

Education

2017 – 2022 Mar Mar **Doctor of Philosophy in Nuclear Physics**, *Australian National University*, Canberra, Research School of Physics and Engineering, Department of Nuclear Physics
Awarded 18 March 2022

Skills and Experience

Accelerator mass spectrometry, nuclear, radiation and space physics

Data analysis, scientific computation, analogue and digital data acquisition electronics

Heavy-ion detection, gas ionisation chamber detectors; γ -ray, conversion electron, electron-positron pair, and neutron spectroscopy

Highlighted Achievements

Invited Seminars: Oak Ridge National Laboratory, Lawrence Livermore National Laboratory, Brookhaven National Laboratory

2020 Jul **Evidence for shape coexistence and superdeformation in ^{24}Mg** , *J. T. H. Dowie et al*, *Physics Letters B* **811**, 135855 (2020)

2022 Apr **Electric monopole transition from the superdeformed band in ^{40}Ca** , *E. Ideguchi, T. Kibédi, J. T. H. Dowie et al*, *Physical Review Letters* **128**, 252501 (2022)

2021 Dec **Implementing Australia's nuclear submarine program**, *Andrew Nicholls, Jackson Dowie and Marcus Hellyer*, *The Strategist*, <https://www.aspistrategist.org.au/implementing-australias-nuclear-submarine-program/>